CS185 Homework 3 Report

Q-Learning and Actor-Critic Algorithms

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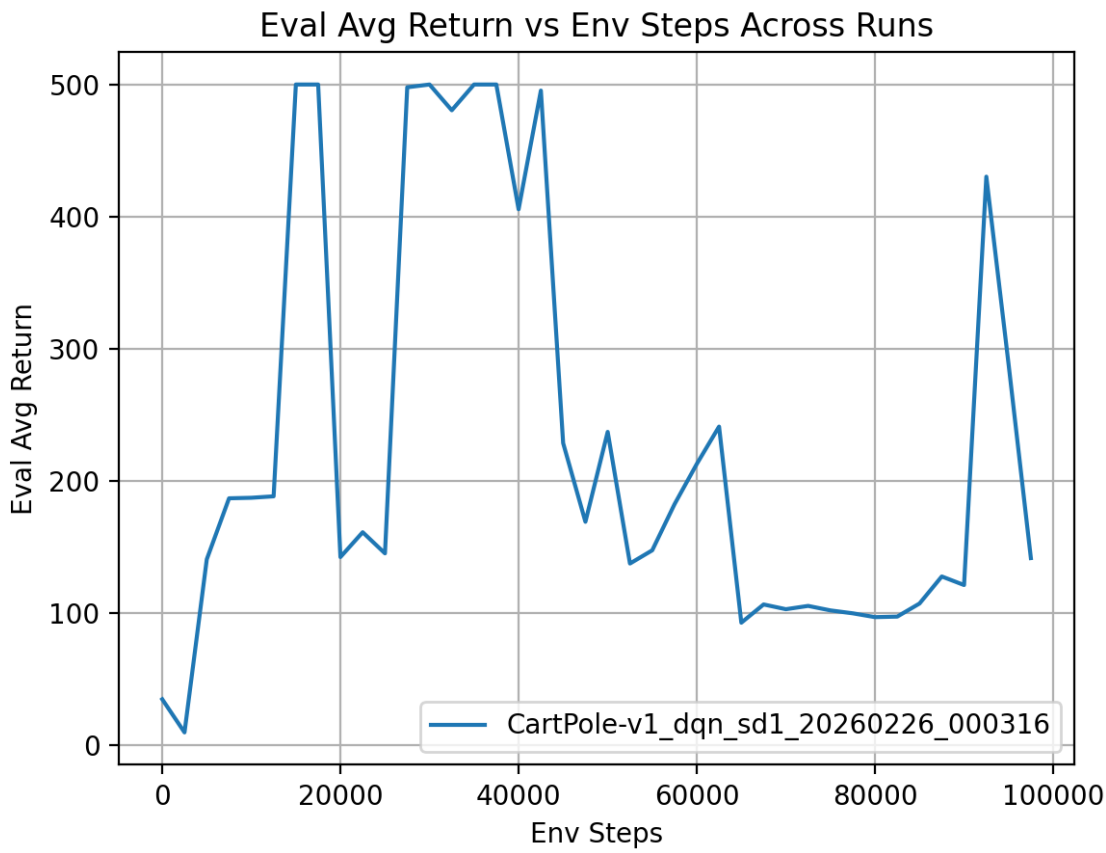
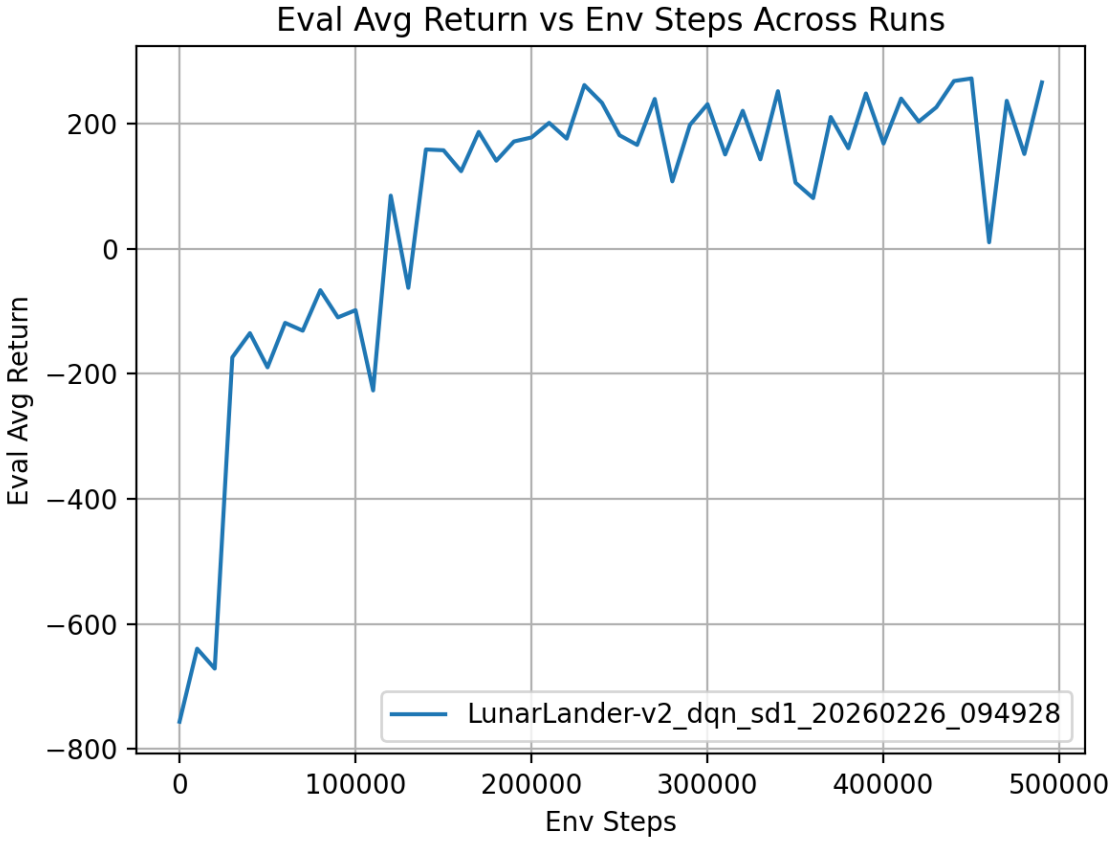
Question 2.4

CartPole-v1 Learning Curve

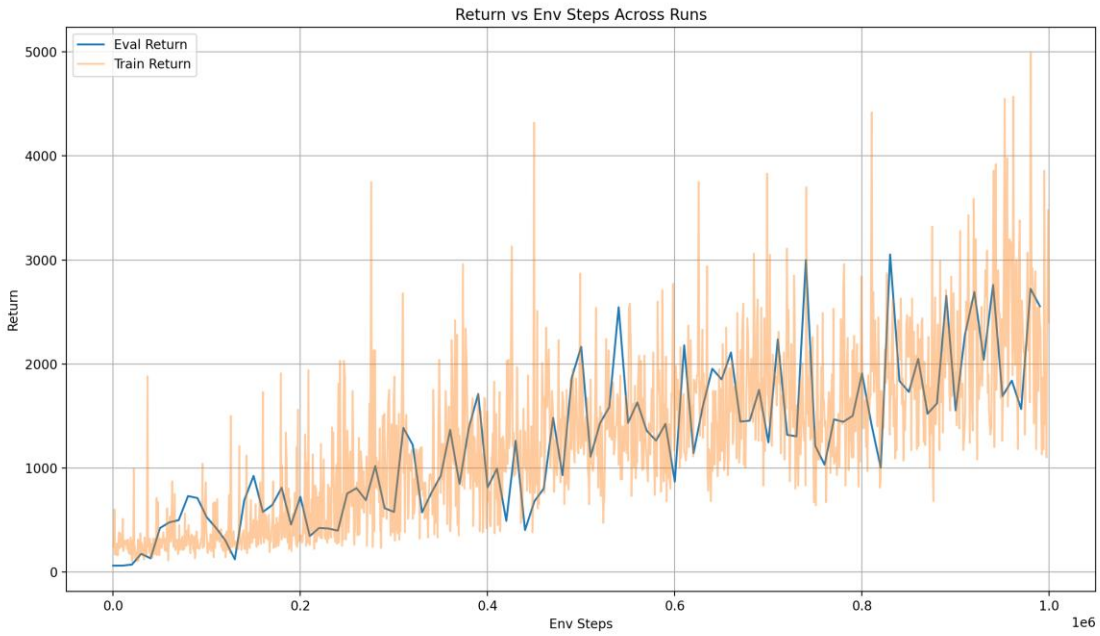


Question 2.5

LunarLander-V2 Learning Curve

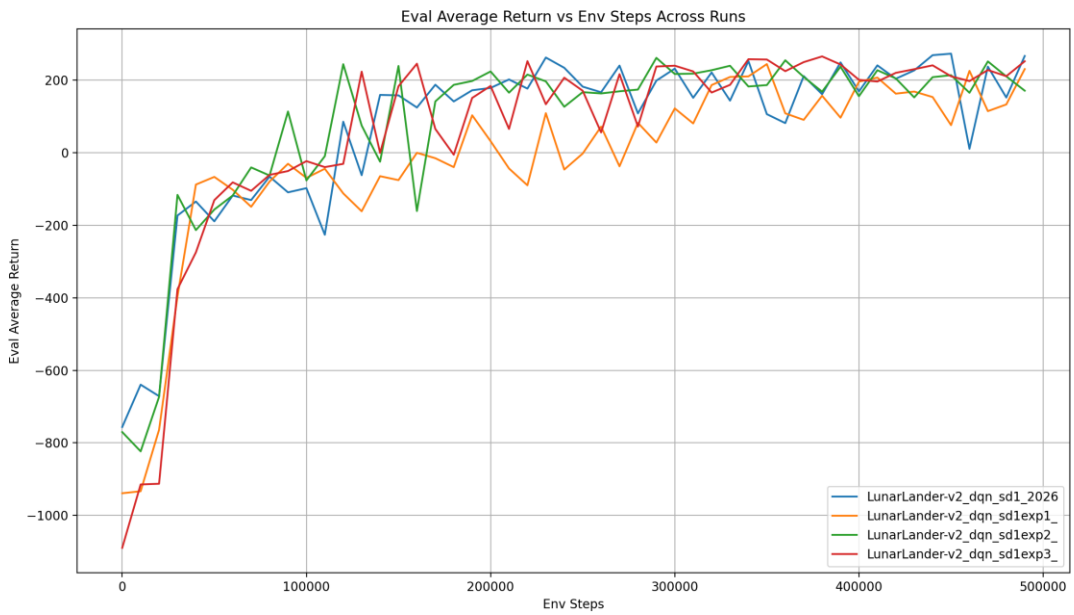


MsPacMan-v0



The difference between eval return and training return early in the training might be due to high initial value of epsilon leading to more aggressive exploration. This would cause actions to be randomly selected which would lead to the train return having high variance. In contrast, the eval return is more stable as the actions taken during evaluation are selected by choosing the action which maximises the online Q function. At later steps, the epsilon decreases as scheduled as the difference between eval and training return look more similar.

Question 2.6



I opted to experiment with the epsilon value which affects the exploration schedule of the policy. I was interested to see how much the rate at which epsilon decays and the value of epsilon would affect the average evaluation return.

Conceptually, my understanding of the original exploration schedule was to let the model explore aggressively over the first 30% of the steps so that it could simulate a variety of trajectories, ideally discovering quality state-action transitions which maximise the return, and provide coverage of the state space. Then exploit these transitions greedily over the remainder of the steps whilst maintaining some degree of exploration in the event the model does not collect a sufficient amount of these transitions in the first 30% of the steps.

Thus, In Experiment 1, I decreased the rate at which the decay of epsilon took place to encourage more exploration in one experiment, experimenting to see if aggressively exploring over more steps would pay off through discovery of more quality transitions. This led to the worst average evaluation return.

In Experiment 2, I increased the rate at which epsilon decayed so that exploration ended early, hoping to see whether the return would increase. This generally led to better returns despite the risk of the policy converging suboptimally since there would be less exploration.

In Experiment 3, I allowed the model to explore throughout the total steps whilst decreasing the value to a lower amount to see if allowing exploration but reducing the probability of it would increase return. This did not lead to much improvement, though it led to better return than experiment 1.

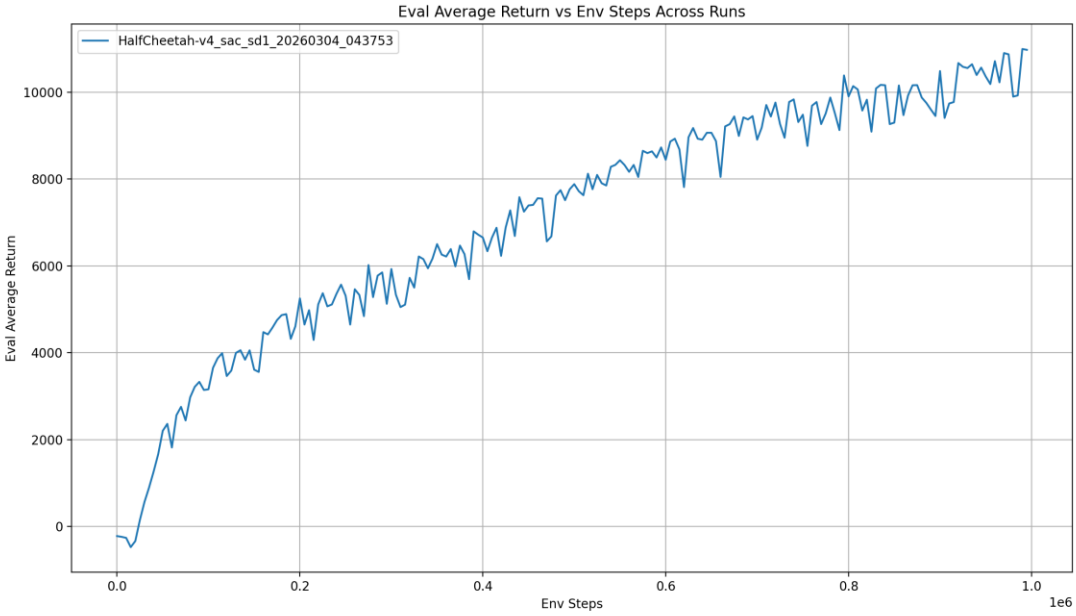
Here are the configurations I used

```
# Original
exploration_schedule = PiecewiseSchedule(
[
(0, 1.0),
(total_steps * 0.3, 0.1),  # Decay to 0.1 over 30% of steps (more gradual)
(total_steps * 0.6, 0.02),  # Then decay to 0.02 by 60% of steps
],
outside_value=0.02,
)
# Experiment 1: more gradual decay
exploration_schedule = PiecewiseSchedule(
[
(0, 1.0),
(total_steps * 0.5, 0.1),  # Decay to 0.1 over 50% of steps (more gradual)
(total_steps * 0.7, 0.01),  # Then decay to 0.01 by 70% of steps
],
outside_value=0.02,
)
# Experiment 2: end exploration early
exploration_schedule = PiecewiseSchedule(
[
(0, 1.0),
(total_steps * 0.2, 0.1),  # Decay to 0.1 over 20% of steps
(total_steps * 0.7, 0.01),  # Then decay to 0.01 by 70% of steps
],
outside_value=0.02,
)
# Experiment 3: Don't end exploration
```

```
exploration_schedule = PiecewiseSchedule(
[
(0, 1.0),
        (total_steps * 0.2, 0.1),  # Decay to 0.1 over 20% of steps
        (total_steps, 0.01),  # Then decay to 0.01 over remaining steps
    ],
    outside_value=0.02,
  )
```

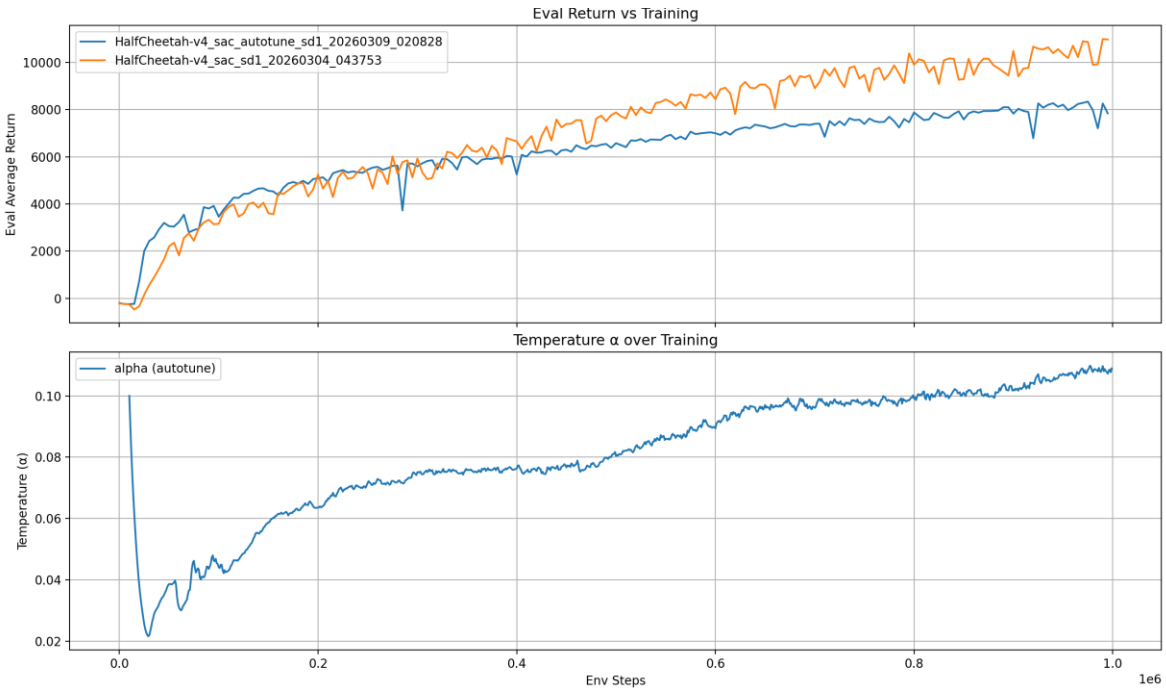
Question 3.4

HalfCheetah Learning Curve



Question 3.5

HalfCheetah vs Autotuned HalfCheetah Learning Curves



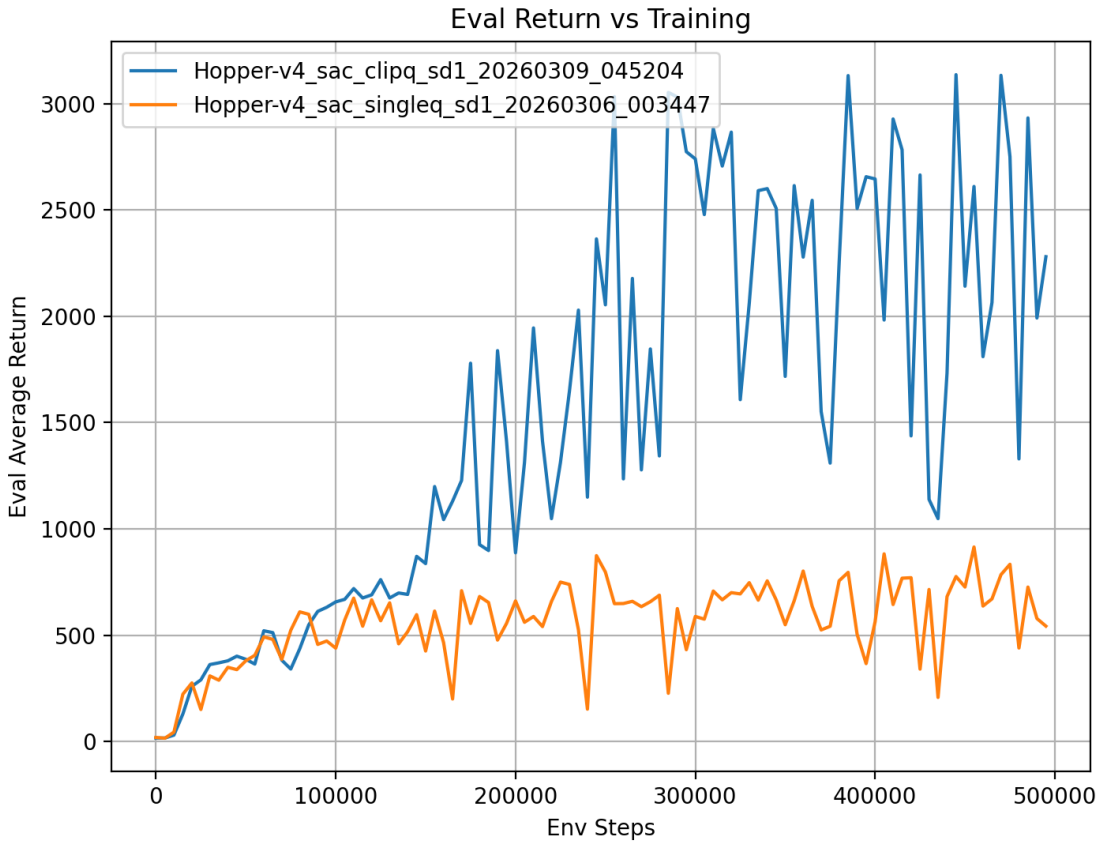
- Does auto-tuning improve or achieve comparable performance to the fixed temperature?

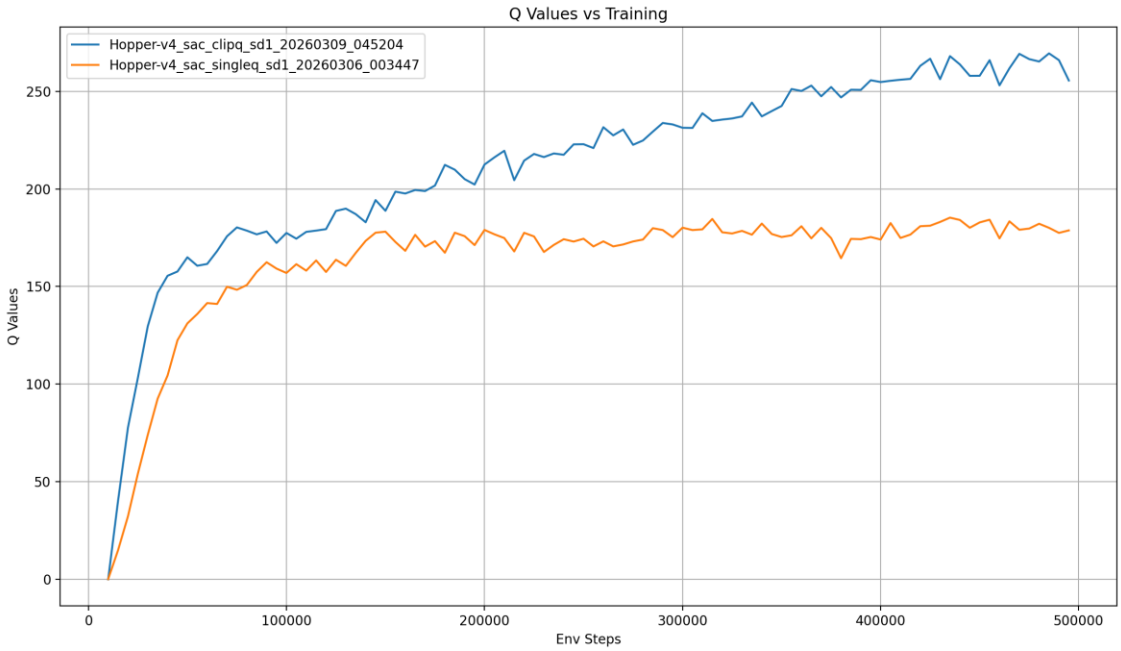
  Auto-tuning improves performance at early steps, but fixed temperature outperforms auto-tuning eventually.

- How does the temperature evolve during training? Does it increase, decrease, or stabilize?
  Temperature decreases sharply to near 0 before increasing and stabilising slightly above its original value.

- Why might the temperature change in this particular way for HalfCheetah?
  Due to high dimensional action space of HalfCheetah, the initial entropy of HalfCheetah might already be higher than the target entrop (-action_dim). Hence, the autotuning would reduce the value of the temperature (acting as a Lagrangian Multiplier) to decrease the entropy bonus since the constraint on the entropy to be greater than target entropy is already satisfied. However, as training progresses and the policy stabilizes, the entropy might decrease, requiring an increase in alpha to maintain the constraint on the target entropy.

Question 3.6


Eval Return vs Training
Hopper-v4_sac_clipq_sd1_20260309_045204
Hopper-v4_sac_singleq_sd1_20260306_003447
Eval Average Return
Env Steps
0
500
1000
1500
2000
2500
3000
0
100000
200000
300000
400000
500000


Q Values vs Training
Hopper-v4_sac_clipq_sd1_20260309_045204
Hopper-v4_sac_singleq_sd1_20260306_003447
Q Values
Env Steps
0
50
100
150
200
250
0
100000
200000
300000
400000
500000

Which one works best? Discuss how these results relate to overestimation bias.

Clipped Q implementation performs better than the single-Q implementation. In single-Q case, the critic might overestimate Q_values which cause the actor to unintentionally select suboptimal actions. The Clipped Q method, on the other hand, involves taking the minimum of the q_values outputted by multiple critics and reduces the overestimation bias leading to more stable learning and hence better performance.